

Project Requirements Document (PRD)

Project Overview

Title: Machine learning based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

Domain: Financial Analytics & Real Estate Market Intelligence

Collaborating Organizations:

- [Unified Mentor](#)
- [Parcl Co. Limited](#)

Background and Context:

Real estate companies operate in markets where buyer behavior is highly diverse:

- Individual home buyers
- Institutional investors
- International buyers
- High-net-worth investors
- First-time buyers

Without segmentation, companies treat all buyers the same.

This leads to:

- Inefficient marketing
- Poor customer targeting
- Missed investment opportunities

Using AI-based clustering, we can discover hidden patterns in buyer behavior.

Problem Statement

Parcl currently lacks a data-driven understanding of:

- Different types of property buyers
- Investment motivations across demographics
- Geographic differences in investment behavior

- Customer financing patterns

This leads to:

- Inefficient marketing spend
- Generic property recommendations
- Poor investor targeting

Dataset Fields Description:

Feature	Description
client_id	Unique client identifier
client_type	Individual / Corporate
gender	Gender of buyer
country	Country of residence
region	Geographic region
date_of_birth	Age indicator
acquisition_purpose	Investment / Personal use
loan_applied	Financing indicator
referral_channel	Source of customer acquisition
satisfaction_score	Customer satisfaction rating

Data Science Methodology (Step-by-Step)

Step 1 – Data Cleaning

Tasks performed:

- Handle missing client attributes
- Normalize categorical labels & Remove duplicate client entries

Step 2 – Feature Encoding

Convert categorical fields using:

- One-Hot Encoding
- Label Encoding

Variables encoded include:

- client_type
 - region
 - acquisition_purpose
 - referral_channel
 - country
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Step 3 – Feature Scaling

Use **StandardScaler** or **MinMaxScaler** to normalize numeric variables such as:

- Age
 - Satisfaction score
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Step 4 – Clustering Model Selection

Two clustering approaches are used.

K-Means Clustering

Advantages:

- Efficient
- Easy to interpret

Hierarchical Clustering

Advantages:

- Reveals nested cluster relationships
- Helps validate K-means results

Step 5 – Optimal Cluster Selection

Use evaluation methods:

Elbow Method



Identifies the optimal number of clusters.

Silhouette Score

Measures clustering quality.

Step 6 – Cluster Interpretation

Each cluster is analyzed based on:

- investment purpose
- geographic distribution
- loan behavior
- customer demographics

Recommended Buyer Segments

Cluster	Buyer Type	Characteristics
C1	Global Investors	High income, investment purchases
C2	First-Time Buyers	Younger, loan dependent
C3	Corporate Buyers	Companies purchasing multiple units
C4	Luxury Investors	High satisfaction, large investments

Streamlit Web Application Requirements

Dashboard Modules

Buyer Segmentation Overview

- Displays cluster distribution.

Investor Behavior Dashboard

- Shows investment patterns by cluster.

Geographic Buyer Analysis

- Maps buyer segments by region.

Segment Insights Panel

- Provides descriptive statistics per cluster.
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User Controls

Users can filter by:

- country
- region
- acquisition purpose
- client type

Deliverables and Submission

- Research paper (EDA, insights, recommendations)
- Streamlit dashboard (live analytics)

Conclusion

This project introduces **AI-driven buyer intelligence** into the Parcl real estate platform. By identifying hidden customer segments through clustering algorithms, the system reveals investment behaviors that traditional analytics cannot detect. Enabling Parcl to build smarter marketing strategies and drive data-driven real estate investment decisions

[Access Dataset](#)